

Satellite Structural Inspection Interferometric Synthetic Aperture Radar (InSAR) Based

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METHODOLOGY

1. Satellite SAR System with Resolution and Revisit Time

To select a suitable SAR system, technical specifications such as spatial resolution, frequency band, polarization, and revisit time were considered. Sentinel-1, operating at C-band (5.405 GHz), was selected due to:

Parameter	Sentinel-1 C-SAR (IW Mode)
Frequency Band	C-band (~5.405 GHz)
Revisit Time	12 days (single satellite)
Spatial Resolution	5 m (range) × 20 m (azimuth)
Coverage Mode	Interferometric Wide Swath (IW)

2. Retrieving Multi-Temporal SAR Imagery Over the Specified Period

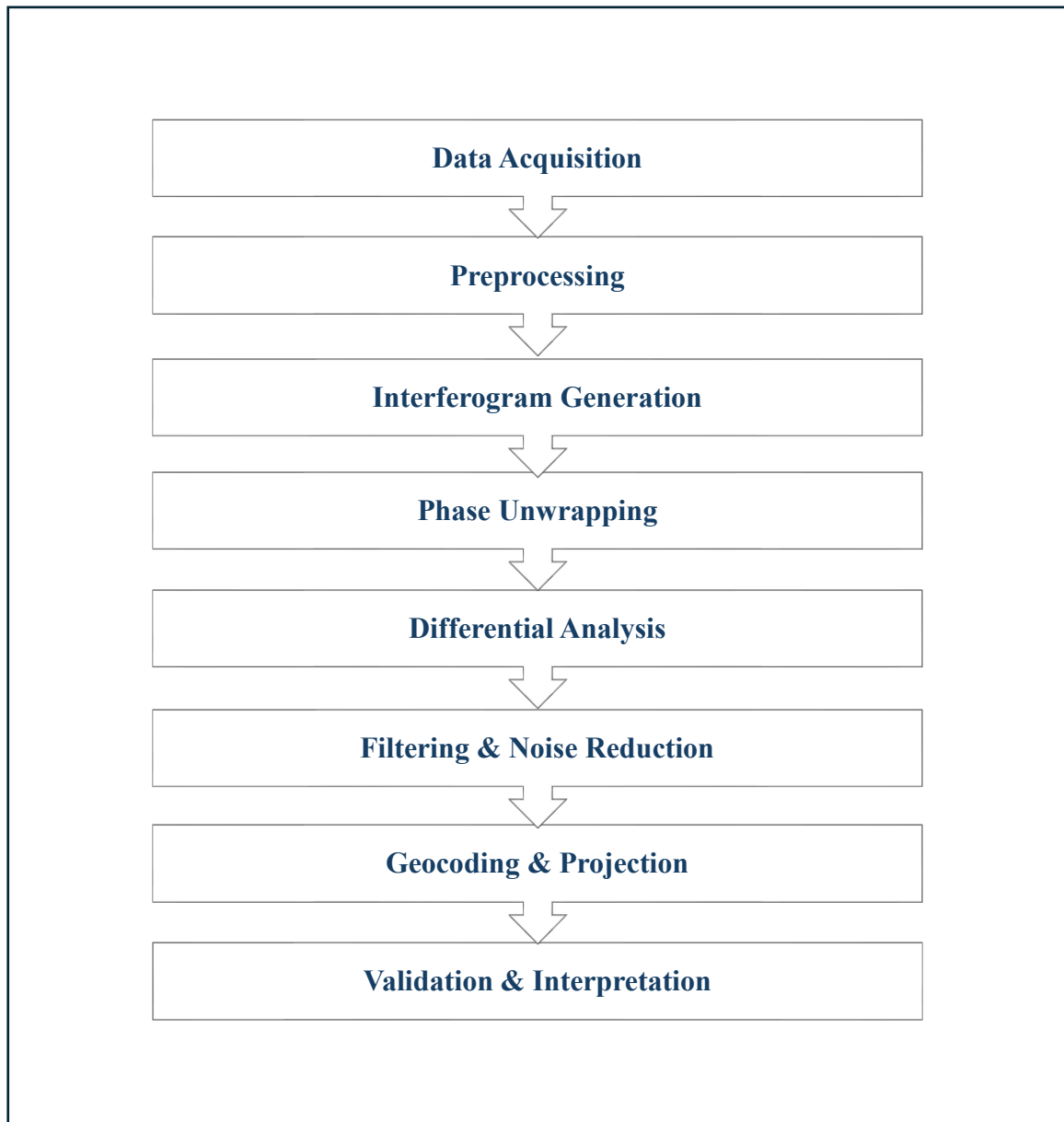
Multi-temporal **Sentinel-1 SLC IW mode** data were downloaded based on:

- **Area of Interest (AOI):** A polygon covering the bridge and surrounding terrain.
- **Date range:** Multiple acquisitions were collected spanning several months to capture temporal variations.
- **Image acquisition conditions:** All images were selected with the same relative orbit, pass direction (ascending or descending), and polarization (typically VV/VH) to ensure consistency for interferometric analysis.

S1A_IW_SLC__1SDV_20250111T130240_20250111T130310_057397_0710AB_813D
S1A_IW_SLC__1SDV_20250123T130239_20250123T130309_057572_07179E_5431
S1A_IW_SLC__1SDV_20250204T130238_20250204T130308_057747_071E86_3FE6
S1A_IW_SLC__1SDV_20250216T130238_20250216T130308_057922_072592_C312
S1A_IW_SLC__1SDV_20250228T130238_20250228T130307_058097_072CB4_7A14
S1A_IW_SLC__1SDV_20250312T130238_20250312T130308_058272_0733CA_217F
S1A_IW_SLC__1SDV_20250324T130238_20250324T130308_058447_073AB5_B445
S1A_IW_SLC__1SDV_20250405T130238_20250405T130308_058622_0741CE_2B52
S1A_IW_SLC__1SDV_20250417T130238_20250417T130308_058797_0748F4_3FFD
S1A_IW_SLC__1SDV_20250429T130239_20250429T130309_058972_07501E_08D8
S1A_IW_SLC__1SDV_20250511T130238_20250511T130308_059147_0756BA_DE13
S1A_IW_SLC__1SDV_20250523T130238_20250523T130308_059322_075CCB_14B2
S1A_IW_SLC__1SDV_20250604T130237_20250604T130307_059497_0762D9_3AB5
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S1A_IW_SLC__1SDV_20250628T130236_20250628T130306_059847_076EE3_6BFD

3. Differential SAR Interferometry (D-InSAR) for Displacement Measurement

D-InSAR analysis was performed to measure millimetre-scale deformation over time:



VISUALIZATIONS

1. Defining specific bridge location & parameters for satellite SAR based inspection



Project & Bridge Details

State – Rajasthan

Bridge No. – 899/1

Bridge Chainage – 943+400

Project - Chittorgarh to Kota

Zone – RJ 06

Structure Type – MNB

Sar Parameters

Sensor – Sentinel-1 (C-Band)

Beam Mode – IW (Interferometric Wide Swath)

Path – 34

Frame – 519

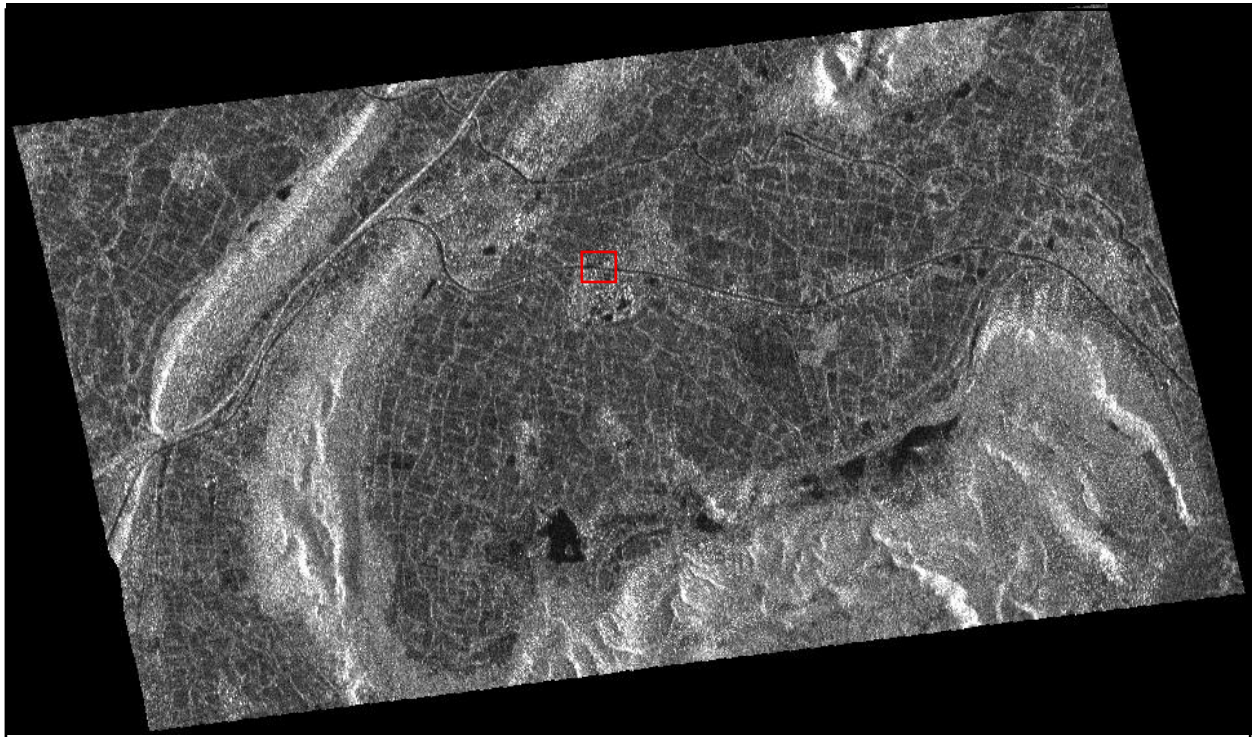
Flight Direction - DESCENDING

Polarization – VV+VH

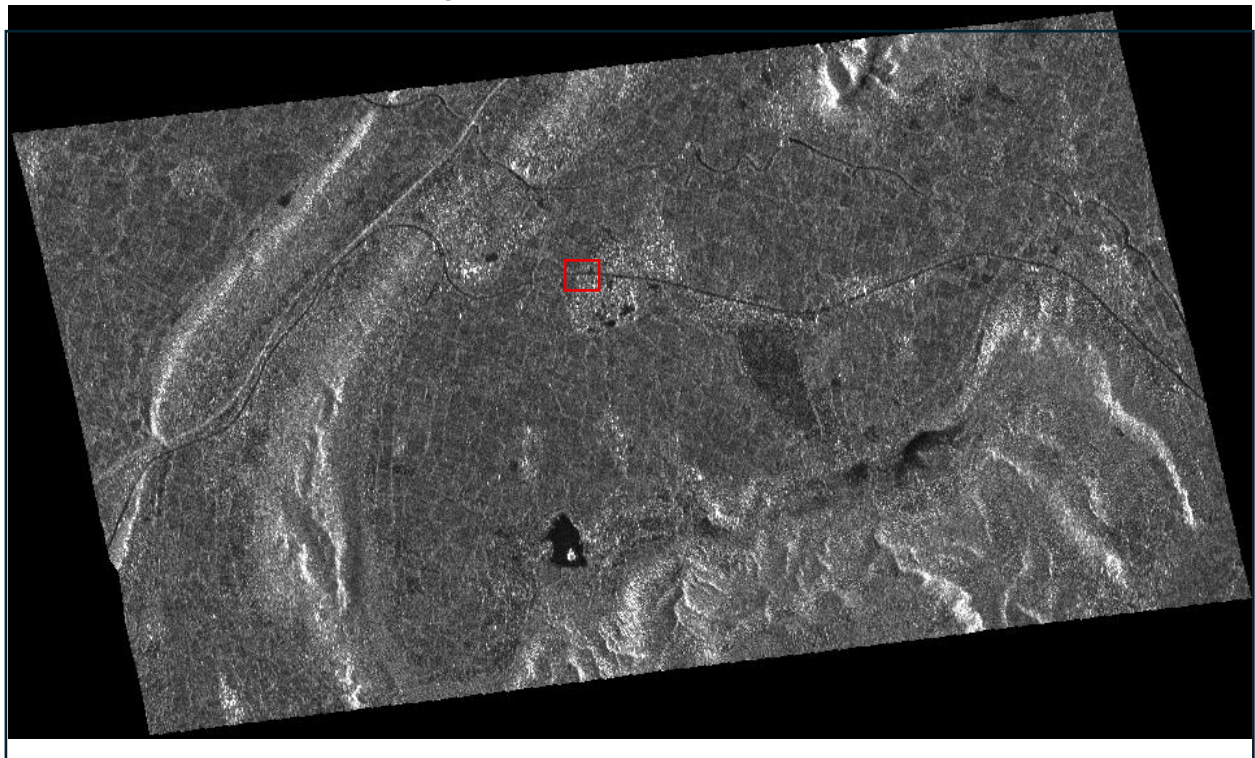
Absolute Orbit – 54706

2. Calibration, georeferencing, orthorectification of SAR images (VH/VV)

- Orthorectification of SAR image VH Polarization



- Orthorectification of SAR image VV Polarization



3. Results of SAR based bridge inspection, including visualizations

Bridge Components

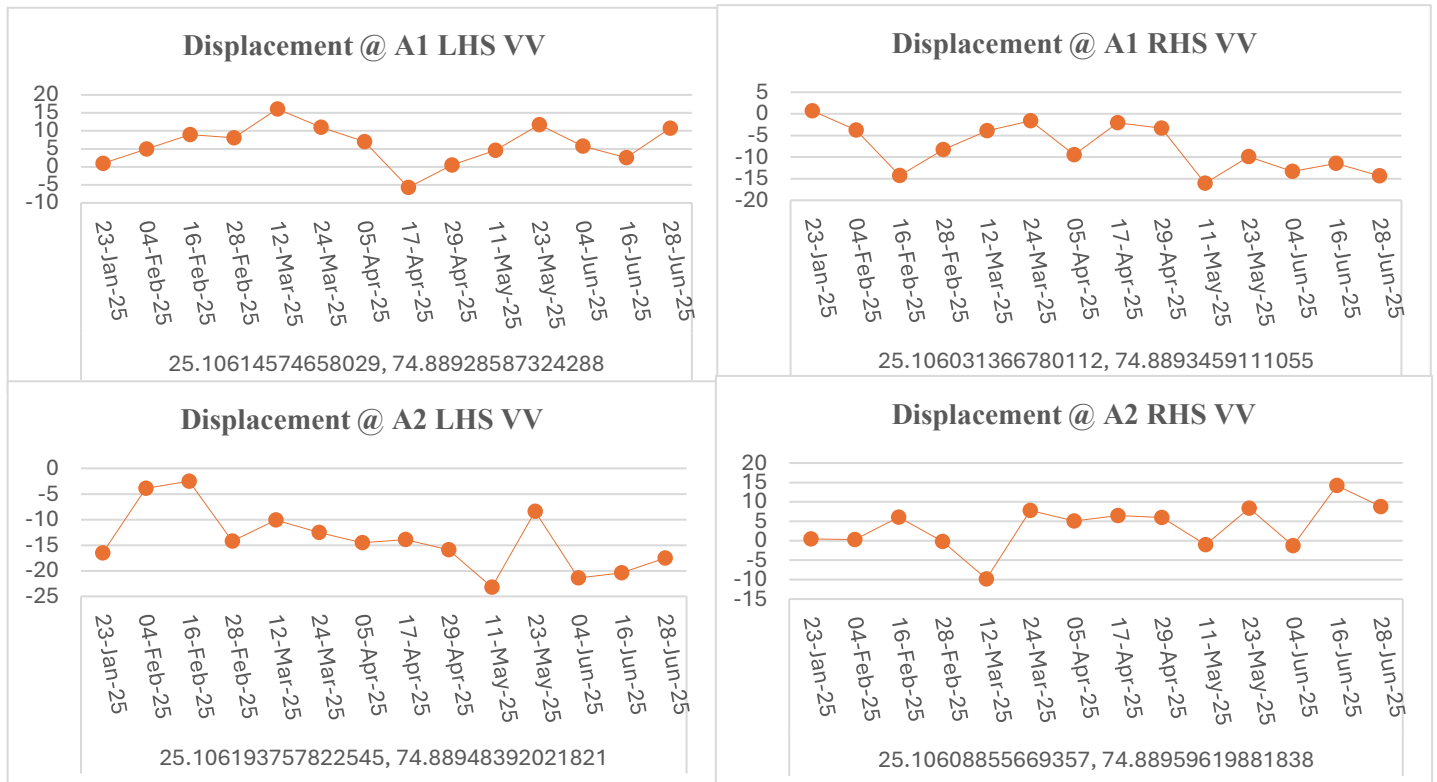
MNB – CH-943+400

Length – 34.72M

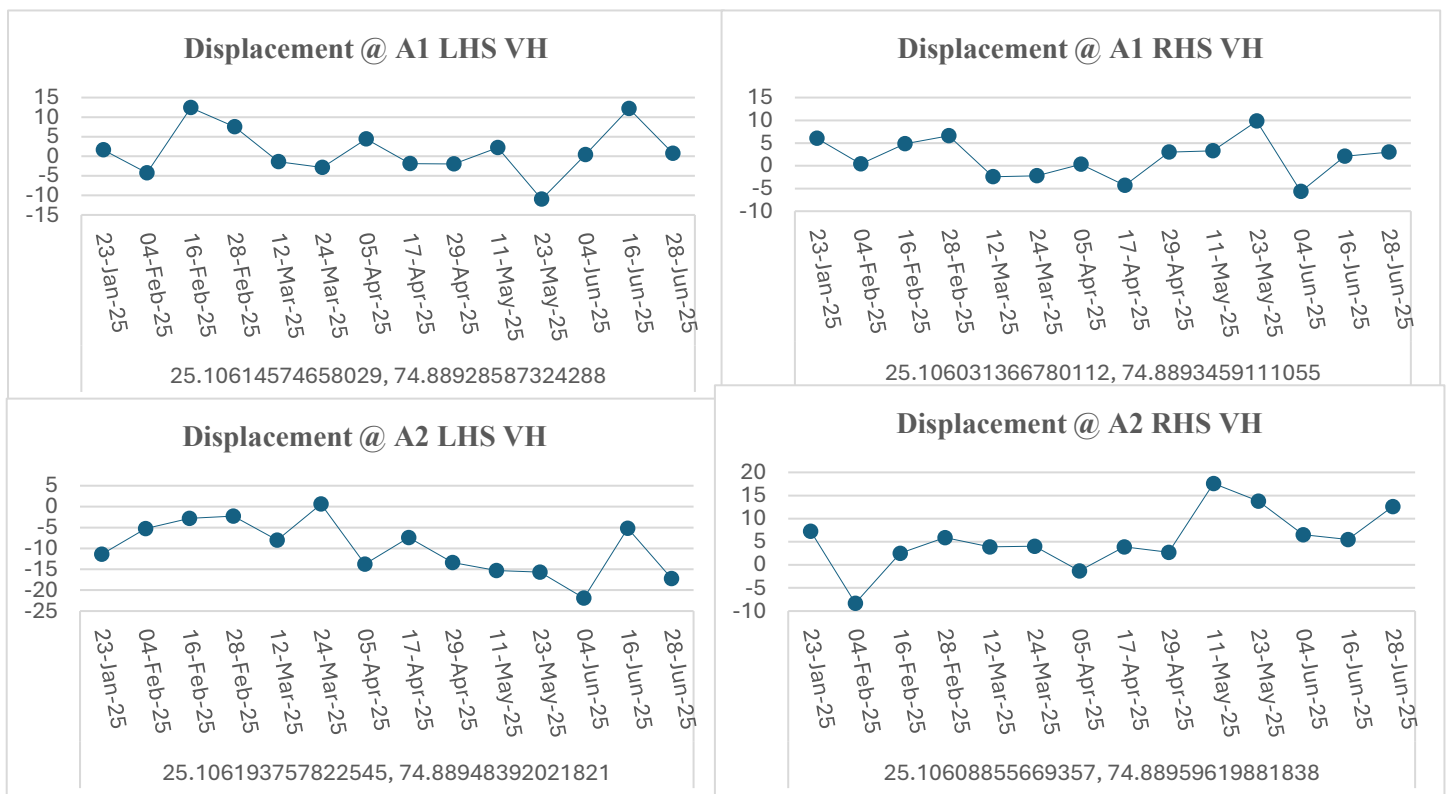


Scan Duration		From : 23 Jan 2025	To : 28 June 2025
Components & Abbreviation			
A1		Abutment 1	
A2		Abutment 2	
LHS		Left Hand Side	
RHS		Right Hand Side	

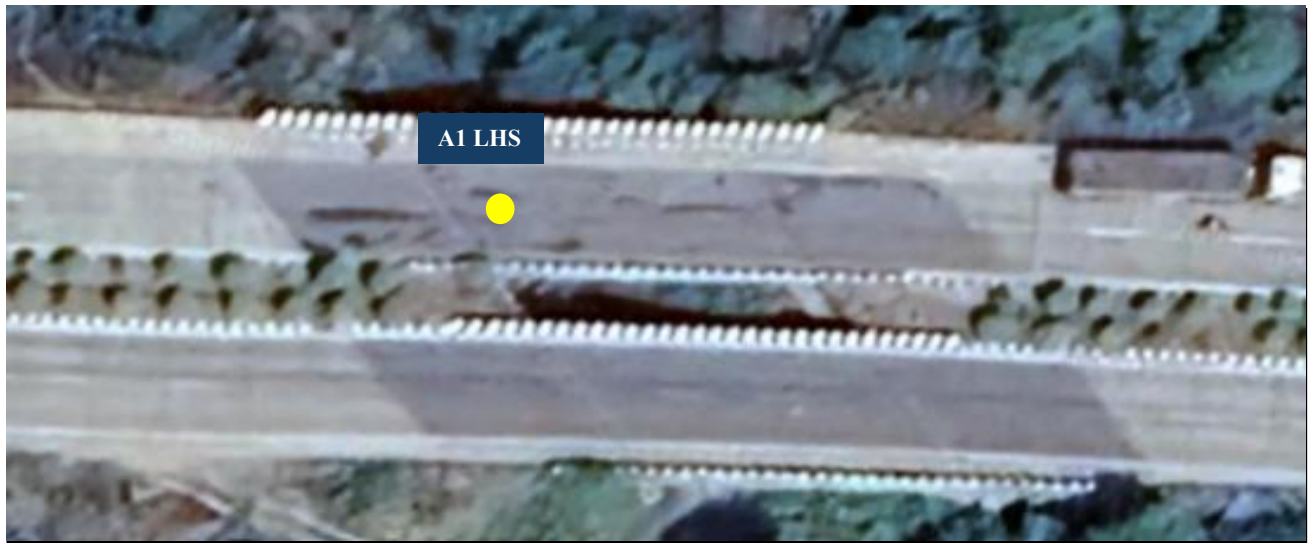
3.1 VV Displacement From 23 Jan 2025 To 28 June 2025



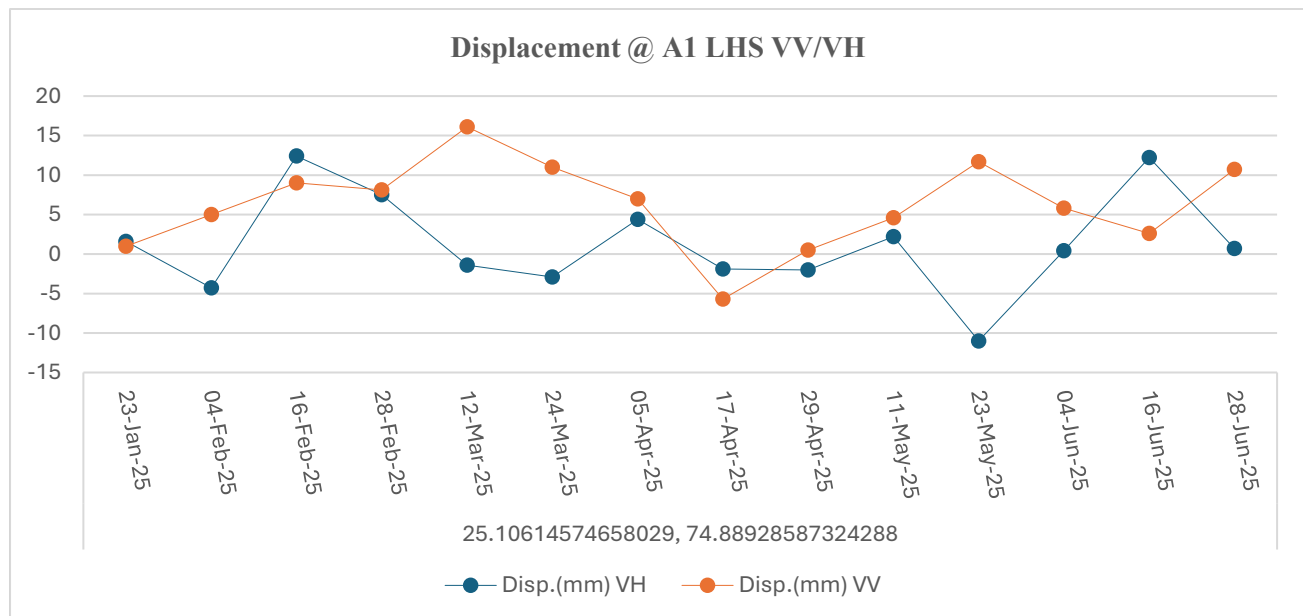
3.2 VH Displacement From 23 Jan 2025 To 28 June 2025



3.3 SAR Based Analysis at Abutment 1 LHS:



A1 LHS (25.10614574658029, 74.88928587324288)



Key Observations:

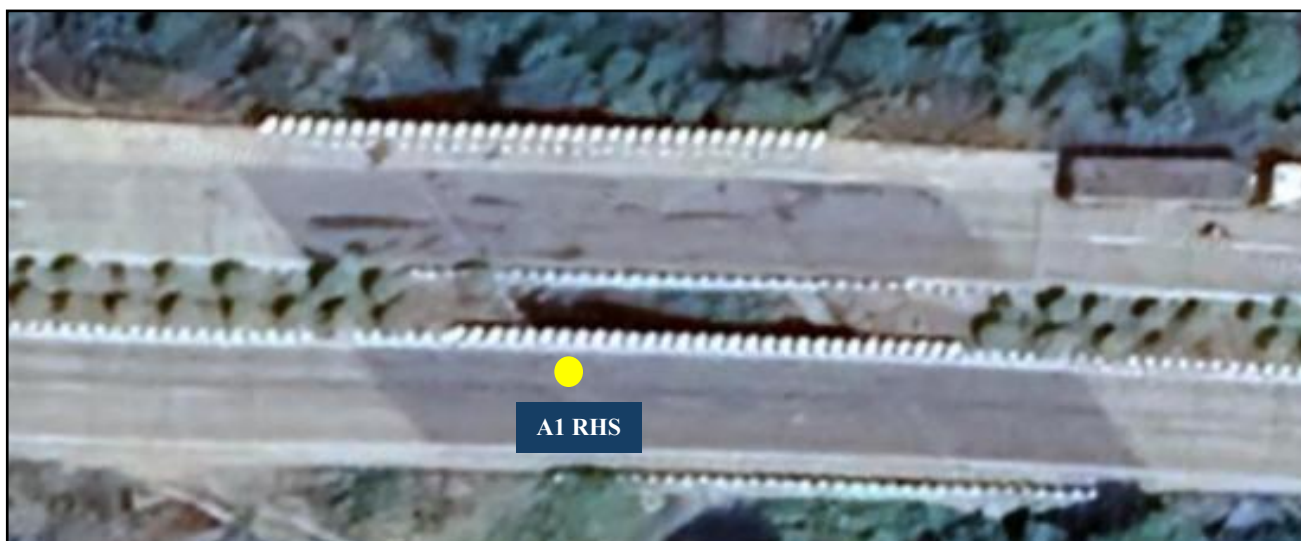
Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** VV varies moderately throughout the monitoring period, generally ranging between –6 mm and +15 mm. Overall, VV shows frequent fluctuations, indicating varying vertical response at the A1 LHS location.
- **Horizontal Displacement (VH)** VH remains mostly within the –12 mm to +12 mm range. Horizontal shifts do not exhibit a long-term monotonic trend but show alternating positive and negative movements, suggesting reversible lateral behaviour.

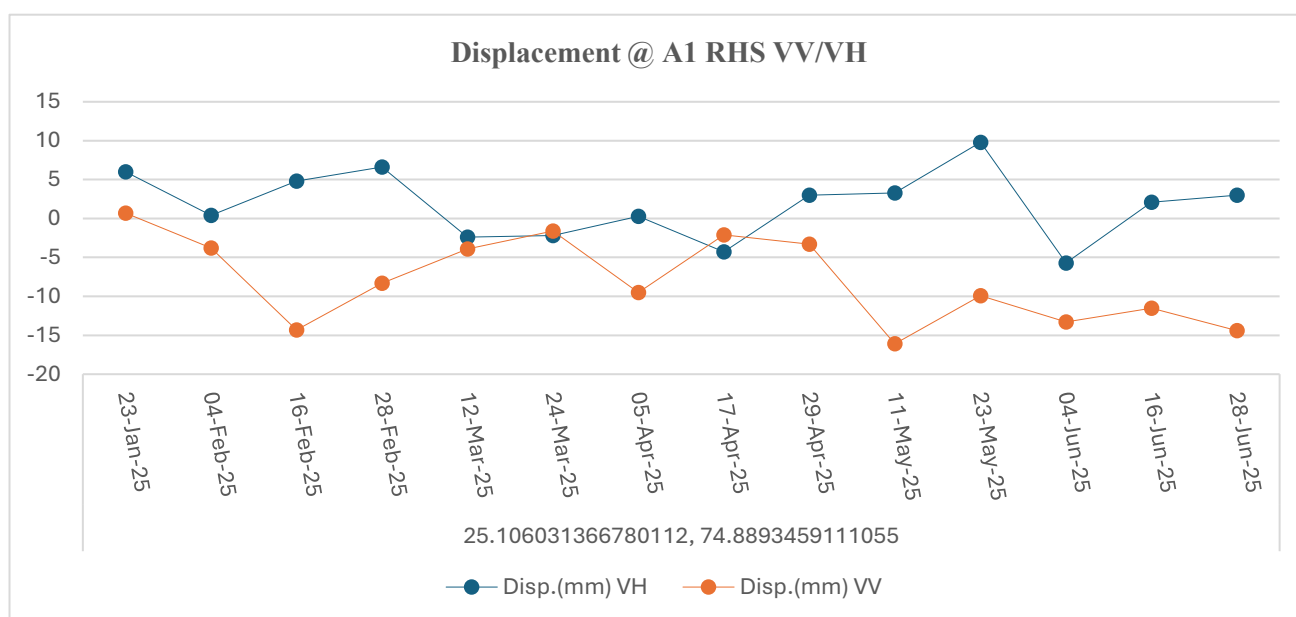
Critical Events & Anomalies:

- **March 2025:** Both VV and VH display significant increases, with VV reaching +15 mm and VH at 4–6 mm, indicating a combined vertical–horizontal movement event.
- **April 2025:** VH shows a sharp downward spike (–12 mm), while VV remains slightly negative, hinting at localized lateral instability.
- **June 2025:** VV increases again to +12 mm, aligning with a moderate positive VH shift, suggesting renewed upward vertical movement.

3.4 SAR Based Analysis at Abutment 1 RHS:



A1 RHS (25.106031366780112, 74.8893459111055)



Key Observations:

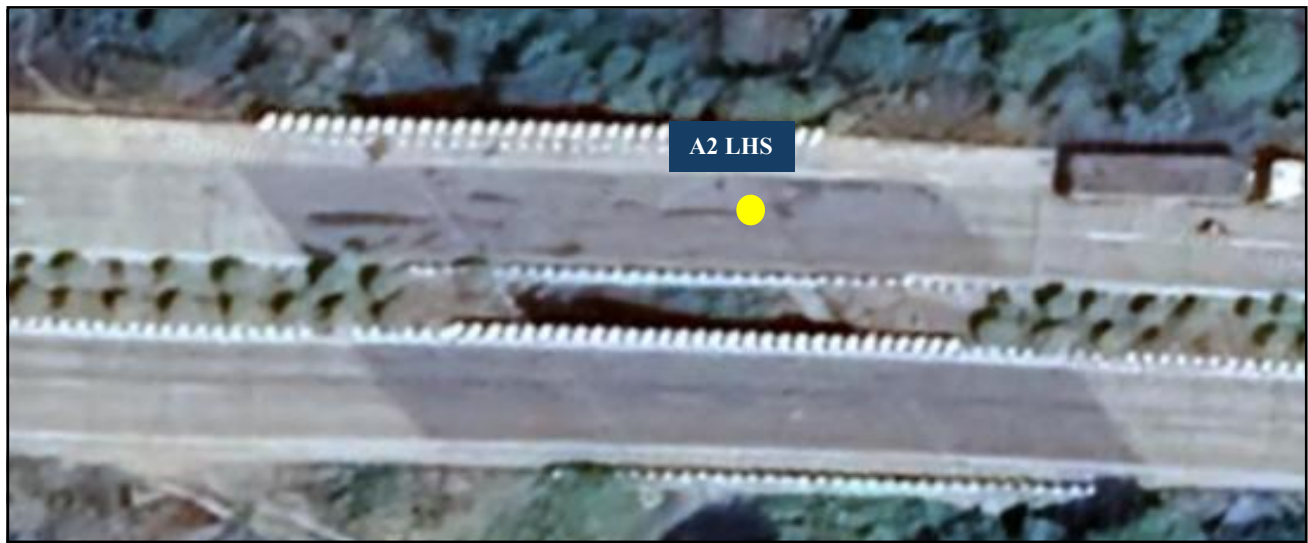
Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** VH exhibits moderate lateral variability, fluctuating between –10 mm (south-west direction) and +10 mm (north-east direction).
- **Vertical displacement (VV)** shows a predominantly downward movement trend, with values ranging from approximately –2 mm to –18 mm across the monitoring period. Periodic peaks of larger downward displacement appear in late February, late April, and late May, indicating recurring vertical settlement phases.

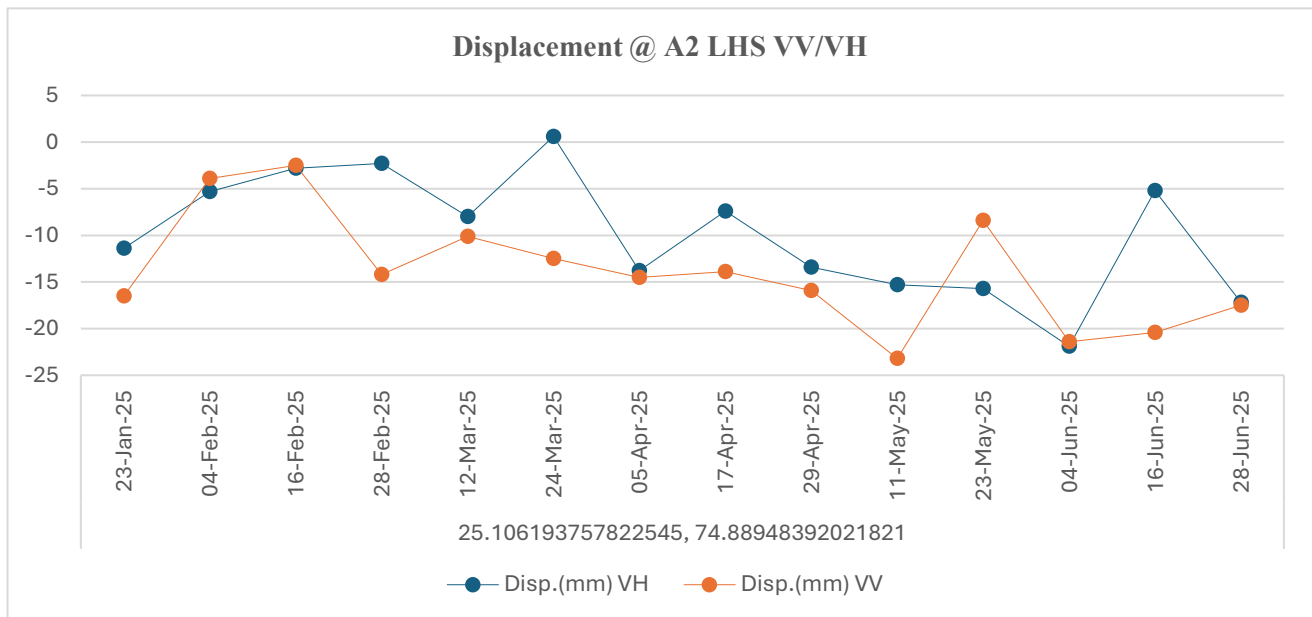
Critical Events & Anomalies:

- **February 2025:** Both VV and VH show pronounced fluctuations, indicating the first notable spike in combined vertical and horizontal activity.
- **May 2025:** VH peaks to nearly +10 mm, while VV again reaches –15 mm, representing another period of significant combined displacement

3.5 SAR Based Analysis at Abutment 2 LHS:



A2 LHS (25.106193757822545, 74.88948392021821)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** shows moderate variability, ranging approximately from **-18 mm to +2 mm**, with several cycles of upward and downward movement, indicating periodic lateral instability on the LHS.
- **Vertical Displacement (VV)** remains consistently negative throughout the monitoring period (**-5 mm to -22 mm**), reflecting a dominant downward trend with no upward recovery, suggesting continuous settlement or downward deformation.

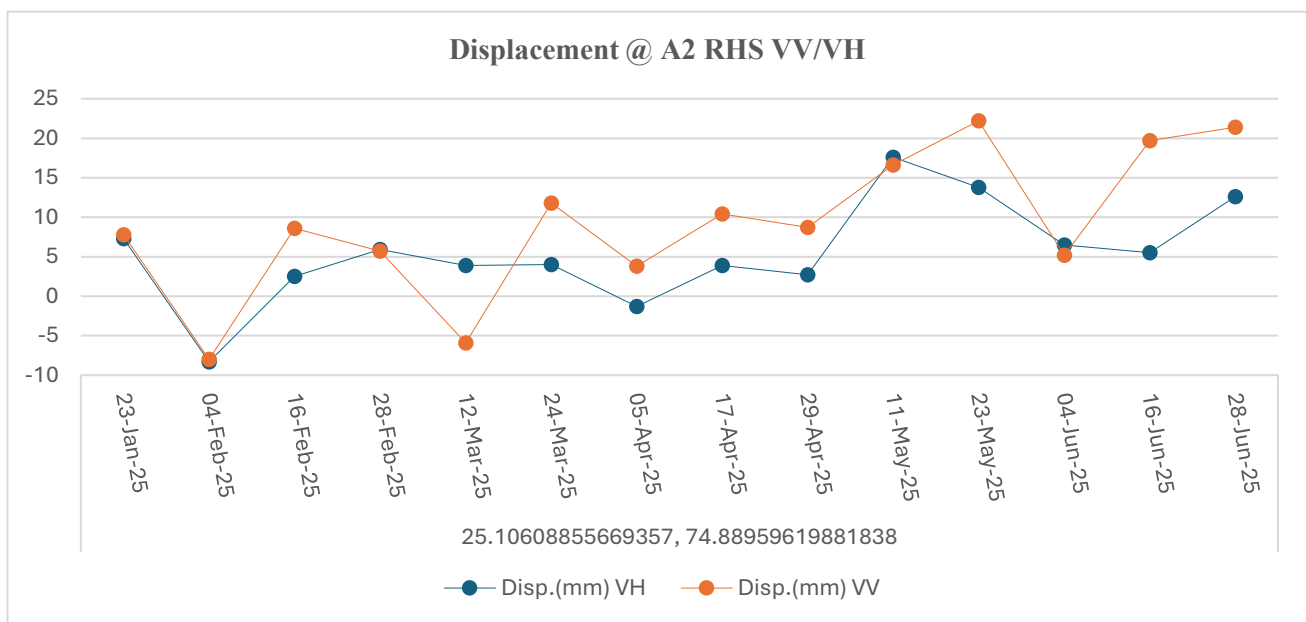
Critical Events & Anomalies:

- **March 2025:** VH experiences a brief positive peak near **+2 mm**, marking the only significant upward shift in an otherwise predominantly negative horizontal displacement pattern.
- **June 2025:** A sharp VH rise from **-15 mm** to nearly **0 mm** signals a temporary lateral correction, while VV remains significantly downward (**≈ -18 mm**), indicating persistent vertical distress.

3.6 SAR Based Analysis at Abutment 2 RHS:



A2 RHS (25.10608855669357, 74.88959619881838)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical (VV) displacement** VV shows a clear upward displacement trend over the monitoring period. Values fluctuate between –8 mm to +22 mm, with progressively increasing peaks observed in March, May, and June 2025, indicating increasing vertical uplift activity at the A2 RHS point.
- **Horizontal (VH) displacement** VH remains comparatively moderate, ranging between –2 mm and +16 mm. The displacement pattern is uneven but shows a gradual rise toward May–June 2025, suggesting increasing lateral activity during the later monitoring stages.

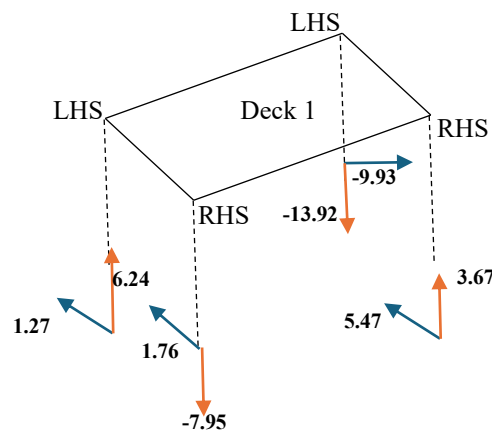
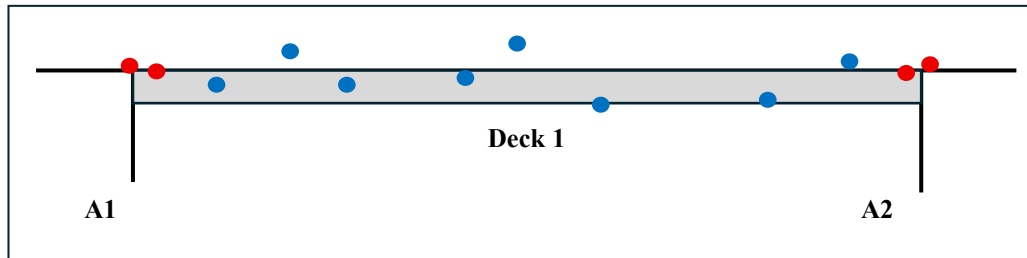
Critical Events & Anomalies:

- **March 2025:** VV jumps sharply to +13 mm, while VH remains stable around 4–5 mm, marking the first noticeable vertical uplift event.
- **June 2025:** VV rises significantly to +18–22 mm, while VH increases to 8–13 mm, showing the strongest combined displacement of the monitoring period.

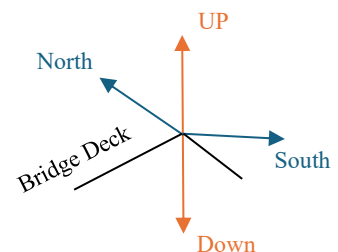
ANALYSIS

1. Identify potential deformation, shifts or movement of the bridge structure

- Points of Analysis (POA) of bridge (●)



Bridge Deck Slab Analysis



Vertical Displacement Behaviour:

- Vertical displacement shows consistent upward movement overall, fluctuating between approximately – 8 mm and +22 mm, indicating progressive vertical rise at the RHS location. A steady upward trend is visible from March to June 2025, with VV increasing from near 0 mm to over +20 mm, suggesting ongoing vertical uplift.

Horizontal Displacement Behaviour:

- Horizontal displacement ranges between –5 mm and +16 mm, indicating moderate lateral variability. The RHS demonstrates a general increase in lateral movement from January to June 2025, with multiple oscillations but an overall upward trend.

CONCLUSION:

- The displacement characteristics at A2 RHS indicate combined vertical uplift and increasing horizontal movement, consistent with progressive deformation rather than random fluctuations. Simultaneous peaks in both VV and VH during May–June 2025 suggest structural actions influencing both vertical and lateral behaviour at the same time—possibly settlement imbalance, structural rotation, or deck-girder interaction.